

GTA V Global Sales & Player Analytics — Problem Statement

1. Background

Grand Theft Auto V (GTA V) is one of the best-selling entertainment products in history, sustaining commercial relevance across more than a decade and multiple console generations. Its revenue model has evolved beyond one-time unit sales to include downloadable content (DLC) and in-game microtransactions (Shark Cards) through GTA Online. With global, transaction-level sales and engagement data available, data analysis can be used to uncover how revenue, player behavior, and monetization have shifted across regions, platforms, and time.

2. Objective

The objective of this project is to analyze a global GTA V sales and player-engagement dataset to identify the key factors driving revenue performance, regional market strength, and the shift in player behavior between Story Mode and GTA Online. The analysis focuses on understanding the impact of geography, platform generation, monetization stream, and promotional events on overall commercial performance using statistical techniques and exploratory data analysis.

3. Dataset Description

The dataset consists of approximately 14,780 transaction-level records spanning 2013–2026, with 47 features including country, platform, sales channel, units sold, gross revenue, discount percentage, DLC and Shark Card revenue, active/online/story-mode player counts, customer ratings, refund rates, and macroeconomic context (GDP per capita, population, internet penetration). A derived **GDP Tier** feature (Low / Medium / High) is engineered from GDP per capita to support economic segmentation analysis.

4. Scope of Analysis

The project includes data profiling, cleaning, feature engineering, exploratory data analysis (EDA), and trend analysis. Techniques such as year-over-year growth analysis, correlation analysis, country/region ranking, and economic-tier segmentation are used to evaluate the relationships between revenue, player engagement, monetization streams, and macroeconomic conditions.

5. Tools & Technologies

The analysis is performed using a combination of tools including Microsoft Excel (pivot tables and stakeholder-facing summaries), SQL (data querying using aggregation, window functions, and CTEs), Python (EDA, correlation analysis, and revenue forecasting with Facebook Prophet), and Power BI (interactive dashboard and visualization).

6. Expected Outcomes

The project aims to identify the strongest drivers of GTA V's commercial performance and provide data-driven insights into regional strength, monetization mix, and the Story Mode-to-Online player shift. These insights support a better understanding of long-tail live-service game economics. The final output includes a structured dataset, a KPI-based analytical report, and an interactive dashboard presenting key findings.